

Guiding generative models to uncover diverse and novel crystals via reinforcement learning

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Hyunsoo Park  & Aron Walsh 

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Discovering functional crystalline materials entails navigating an immense combinatorial design space. Although recent advances in generative artificial intelligence have enabled the sampling of chemically plausible compositions and structures, a fundamental challenge remains: the objective misalignment between the likelihood-based sampling in generative modelling and the targeted focus on underexplored regions where novel compounds reside. Here we introduce a reinforcement learning framework that guides latent denoising diffusion models in finding diverse and novel, yet thermodynamically viable, crystalline compounds. Our approach integrates group-relative policy optimization with verifiable, multi-objective rewards that jointly balance creativity, stability and diversity. Beyond de novo generation, we demonstrate enhanced property-guided design that preserves chemical validity while targeting desired functional properties. This approach establishes a modular foundation for controllable AI-driven inverse design that addresses the novelty–validity trade-off across the scientific discovery applications of generative models.

The discovery of crystalline materials has historically shaped technological revolutions, from semiconductors¹ and battery materials² to catalysts for energy and environmental applications³. Yet, the design space of crystalline compounds is vast and combinatorially complex, with the overwhelming majority of it remaining unexplored⁴. The emergence of artificial intelligence (AI) marks a substantial shift in materials design, as it offers the ability to navigate the high-dimensional composition–structure–property landscape with unprecedented efficiency and enables a systematic exploration that extends far beyond the reach of conventional methodologies^{5,6}.

Recent progress in generative modelling has demonstrated the ability to sample chemically plausible crystalline compounds⁷, with approaches spanning variational autoencoders (VAEs)^{8,9}, generative adversarial networks^{10,11}, autoregressive transformer models^{12–14} and continuous-time approaches, such as score-based (diffusion) models^{5,15,16} and flow matching^{17,18}. Despite these advances, a fundamental misalignment exists between materials discovery and generative modelling. Generative models are typically trained to maximize likelihood^{19,20}, and they learn to reproduce the statistical patterns in the training data. Conversely, the goal of materials discovery is to

navigate systematically towards regions where chemical novelty and thermodynamic viability converge. These are precisely the under-explored spaces, often sparsely represented in existing databases, where viable yet unprecedented compounds will most probably reside. This objective mismatch constrains the capacity of likelihood-based approaches to systematically explore chemical spaces with the greatest scientific interest.

Overcoming this limitation requires generative models that can explore chemical space guided by explicit design objectives rather than statistical similarity to known data. Reinforcement learning (RL) provides a principled framework for realizing this goal, as it optimizes expected rewards instead of the likelihood of the data. Post-training with RL algorithms, such as proximal policy optimization²¹, has enabled generative models to align their outputs with predefined objectives and exhibit controllable behaviours across diverse domains, most notably in language²² and vision²³. Pioneering efforts have demonstrated the potential of RL through fine-tuning autoregressive transformers²⁴, establishing standardized benchmarks²⁵ and guiding diffusion processes in crystal atomic representation²⁶. However, previous applications of RL to crystalline compounds have primarily

operated directly on the crystal atomic representation, which comprises atom types (A), atomic coordinates (X) and lattice matrix (L). However, applying RL within this atomic representation remains challenging. The optimization process is hindered by the hybrid nature of the design space, which necessitates the simultaneous handling of discrete variables (atom types) and continuous variables (the lattice matrix and fractional coordinates). Moreover, this heterogeneity requires distinct diffusion processes for each component, which results in unstable gradient estimation and computational inefficiency during RL training.

Although previous applications of RL to crystalline compounds have primarily operated directly on the atomic representation, our approach leverages a compressed, continuous, latent representation. Thus, we introduce Chameleon2, a policy-gradient RL framework tailored for denoising diffusion models in this continuous latent space. Regarding the generative backbone without RL, although our previous model, Chameleon1⁶, operates as a denoising diffusion model directly on the atomic representation, the base model of Chameleon2 is constructed as a latent denoising diffusion model using a compressed, continuous, latent space. Building on this latent framework, Chameleon2 adopts the group-relative policy optimization (GRPO) algorithm²⁷, which performs group-normalized, preference-based updates to reduce the gradient variance, and it integrates naturally with batched candidate generation to ensure stable optimization. Exploration beyond the empirical data distribution is encouraged through a multi-objective reward design that jointly promotes creativity, validity and diversity, thus effectively addressing the trade-off between novelty and stability that constrains conventional generative models. We benchmarked Chameleon2 against representative baselines, including the latent denoising all-atom diffusion transformer (ADiT)²⁸ and atomic representation diffusion models such as DiffCSP¹⁶, MatterGen⁵ and Chameleon1 (excluding its text encoder, unlike the original framework, which aligned a text encoder with textual descriptions via contrastive learning). By enabling property-guided control without relying on heuristic mechanisms, such as classifier-free guidance (CFG)²⁹, our RL framework provides a principled pathway towards more controllable generation. Furthermore, the modular design of the frameworks allows seamless alignment with diverse reward signals, thus establishing a versatile foundation for the AI-guided inverse design of functional crystalline materials.

Results

RL framework for exploring materials space

Effective computational materials discovery strategies must not just produce plausible structures absent from databases but systematically access underexplored regions of materials space. Current generative AI models excel at interpolative novelty to create plausible structures that statistically resemble the training data. However, they face issues when extrapolating towards unfamiliar chemistries characterized by rare element combinations, unconventional coordination environments or low-probability structural motifs. To address this challenge, we propose an RL framework that guides latent denoising diffusion models, as illustrated in Fig. 1a.

Our framework integrates policy optimization with generative models for crystalline compounds through two interconnected components. The agent comprises a latent denoising diffusion model, which serves as the policy network, and the environment encompasses reward evaluation mechanisms that assess generated structures across several objective functions. This architecture establishes a feedback loop whereby the generative model iteratively refines its exploration strategy based on explicit reward signals.

Before RL training, the agent undergoes pretraining on large structure databases so that it learns essential crystallographic constraints, including local coordination patterns, periodicity and symmetry-related geometries, that characterize physically plausible

compounds. During RL training, the agent updates parameters through on-policy-gradient methods rather than score-matching objectives, so that it transitions from likelihood-based generation to reward-driven exploration.

Latent diffusion model for the policy network

Crystal structures are represented by atom types, coordinates and lattice matrix (A, X, L). Defining a tractable and computationally efficient action space that allows the direct manipulation of these components poses major challenges to the application of RL. Operating directly in the (A, X, L) space renders the log-probability of the policy intractable in policy-gradient methods due to the joint updating of interdependent representations (L and X) and to the mixing of categorical variables (A) with continuous variables (L and X). Furthermore, generating the batches of samples (roll-outs) required for RL exploration becomes prohibitively expensive when operating on high-dimensional real-space representation. Supplementary Fig. 1 indicates that the latent-based Chameleon2 achieves 3.8 times and 5.9 times higher sampling throughput than Chameleon1 (batch sizes of 320 and 1,000), even with Chameleon1 optimized to use only 256 diffusion time steps.

To address these complexities, we adopt a latent diffusion model. Our approach is inspired by the ADiT architecture²⁸, in which crystal structures are first encoded into a unified latent representation $\mathbf{z}$ using VAEs. VAEs create latent representations that facilitate the capture of the coupled relations among composition, geometry and atomic arrangements. This encoding strategy transforms the heterogeneous structure representation into a homogeneous continuous vector in a regularized latent space. This transformation renders the action space tractable for policy-gradient optimization while substantially reducing the computational burden of the denoising process. Within this latent geometry, each denoising step corresponds to an action in which the policy network predicts a noise component that updates the latent state along a reverse-diffusion trajectory. The decoder network subsequently reconstructs full crystal structures from these latent states. Accordingly, the latent formulation is motivated primarily by practical optimization considerations, including tractable policy-gradient updates and reduced roll-out costs. Analogous RL objectives could, in principle, be applied to real-space diffusion models, but the continuous latent space provides an efficient setting for the roll-out-intensive optimization studied here.

Reward design to guide the discovery of new compounds

Effective reward engineering is fundamental to successful RL in materials discovery. The core challenge lies in designing reward signals that are simultaneously verifiable, grounded in physically meaningful quantities and computationally efficient, thus enabling rapid evaluation during the iterative policy optimization.

To achieve a core objective—systematically exploring materials space—we design multi-objective reward functions that balance three critical dimensions: creativity, stability and diversity. First, the creativity reward encourages the generation of chemically distinct structures by evaluating uniqueness and novelty. Unlike binary metrics, we employ a continuous formulation based on average minimum distance (AMD)³⁰ to implement a smooth gradient for optimization, as illustrated in Fig. 1b. Second, the stability reward ensures thermodynamic stability, which we operationally define in this work using the 0 K energy above the convex hull as a practical proxy, without explicitly including finite-temperature or kinetic contributions. We use pretrained machine learning force fields (MLFFs) as a computationally efficient proxy for density functional theory (DFT) calculations to estimate the energy above the convex hull (Fig. 1c), which enables a rapid on-policy evaluation. Finally, to prevent mode collapse and ensure broad coverage of the chemical space, we introduce a diversity reward based on marginal-utility decomposition. This metric assigns high scores to

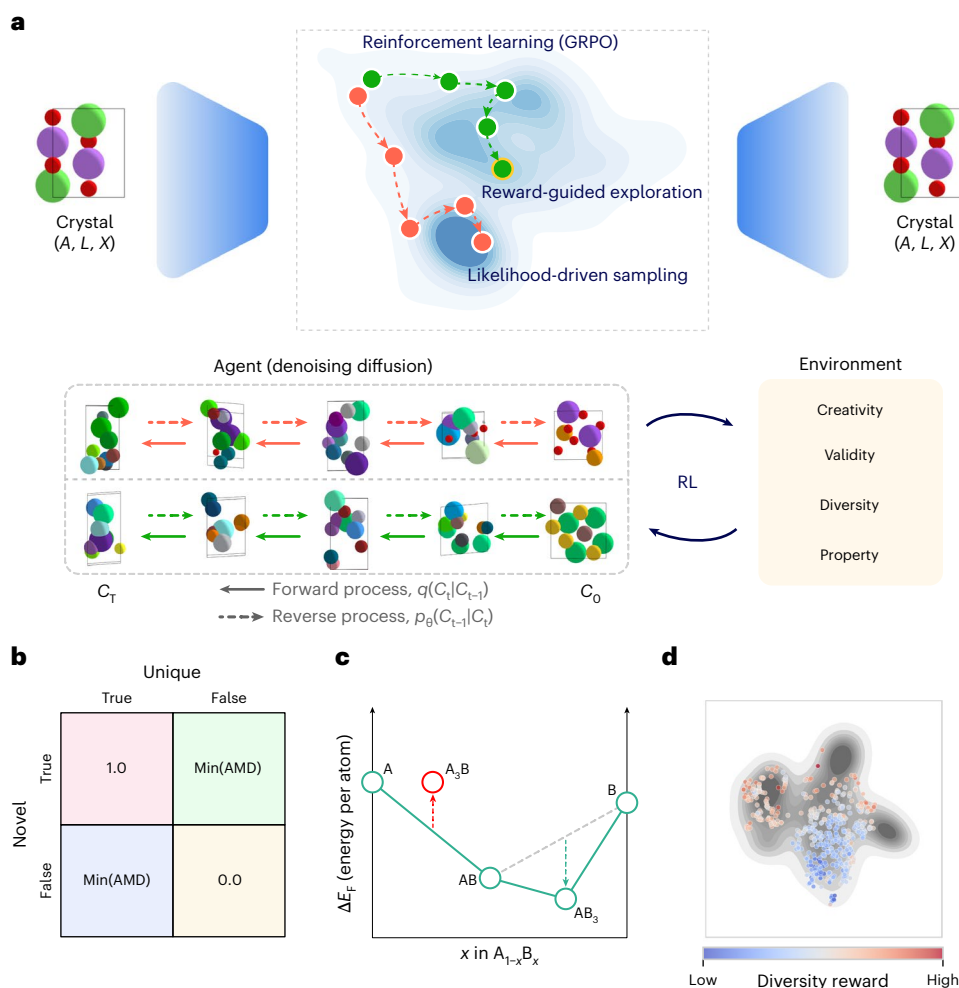


Fig. 1 | RL learning framework for the guided exploration of crystalline materials. **a**, Schematic of the latent diffusion framework with GRPO. The process shifts from standard likelihood-driven sampling (red path) to reward-guided exploration (green path), which enables the agent (denoising diffusion models) to navigate the latent space towards underexplored regions via flexible, multi-objective optimization. **b**, Continuous creativity reward matrix based on AMD, which is designed to penalize known or redundant structures while maximizing rewards for unique and new compounds. **c**, Stability reward mapping

as a function of the predicted energy above the convex hull. The function guides the agent towards thermodynamically stable phases while penalizing unstable configurations. **d**, *t*-SNE projection of structural embeddings showing 1,000 reference (training) structures (grey density contours) and 500 generated structures coloured by the diversity reward. Here, ΔE_f is the formation-energy difference; C_0 , C_i and C_T denote the initial, intermediate and terminal crystal states, respectively; q is the forward process; and p_θ is the learned reverse process.

structures that expand the explored landscape into undersampled regions, as visualized by the *t*-distributed stochastic neighbour embedding (*t*-SNE) projection of the structural embedding space in Fig. 1d. Detailed mathematical formulations and implementation specifics for each component are provided in Methods.

Policy optimization algorithm

To enhance learning stability and reduce gradient variance without the architectural complexity of a separate value network, we adopted GRPO. As illustrated in Fig. 2a, GRPO uses a group-wise advantage normalization strategy such that several candidate structures are sampled and evaluated for each input condition (for example, number of atoms). This approach confines the advantage computation to within subgroups, thus effectively isolating complexity factors. Consequently, as shown in Fig. 2b, GRPO demonstrates lower variance in gradient estimates compared with the standard REINFORCE algorithm^{31,32}, leading to more stable training dynamics. Furthermore, our ablation study underscores the critical role of the diversity reward within this framework. The exclusion of the diversity reward leads to severe mode collapse because the policy exploits local optima through

trivial geometric variations, as reflected in the reward decomposition in Fig. 2c. A detailed analysis of the ablation results is provided in Supplementary Note 1 and Supplementary Fig. 2.

De novo generation benchmark

We benchmarked model performance using a de novo generation task comprising 10,000 structures sampled per model. Crucially, to assess inherent generative capability, all structures were evaluated as-generated without geometry optimization. Figure 3 presents benchmark results for models trained on the large Alex-MP-20 dataset. An analogous analysis on the standard MP-20 dataset is provided in Supplementary Fig. 3 and Supplementary Table 1. We use mSUN (meta-stable, unique and novel) as the principal metric, alongside coverage. Detailed definitions of validity (compositional and structural) and coverage metrics are provided in Methods.

A comprehensive analysis of models trained on the Alex-MP-20 dataset revealed clear performance variations across generative architectures (Fig. 3). Among baseline models, MatterGen achieved the highest mSUN (41.0%), slightly exceeding its originally reported value of 38.6%, which used a workflow of DFT energy evaluations after

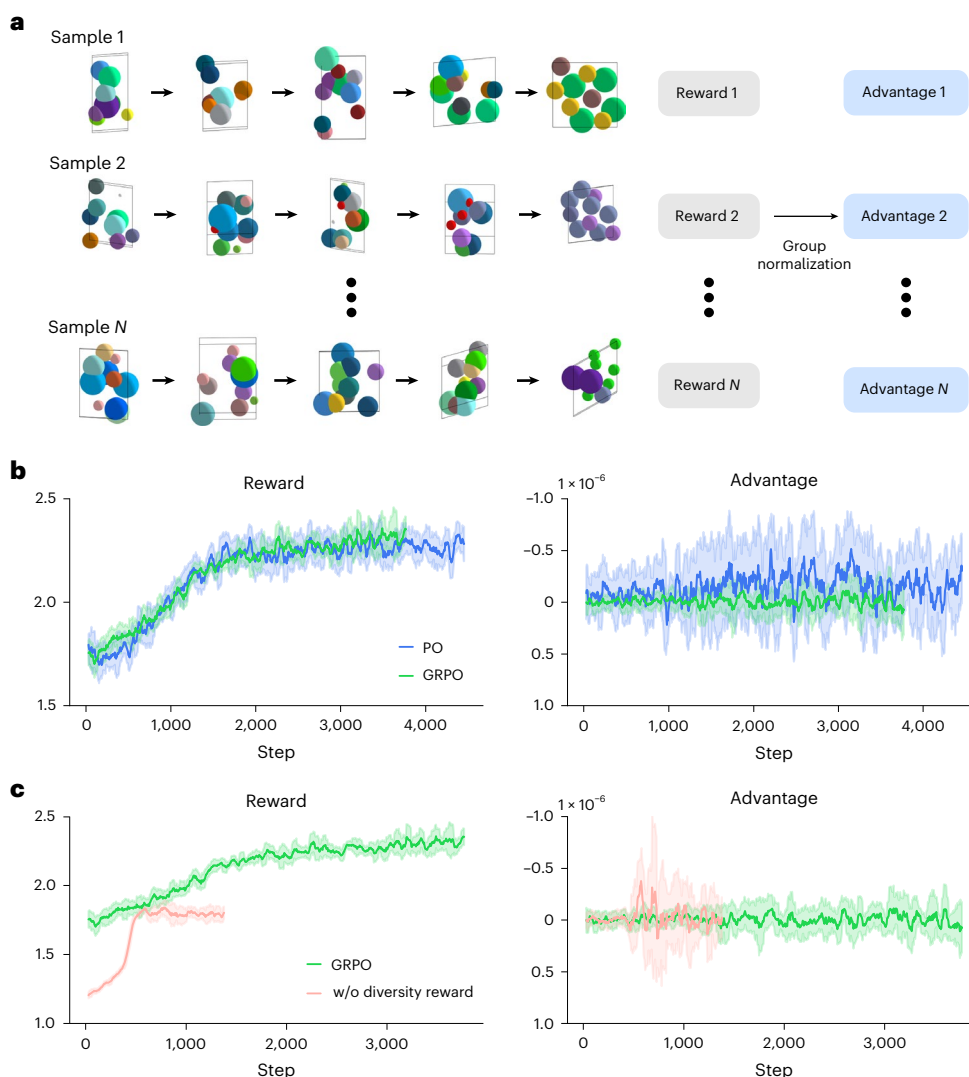


Fig. 2 | Illustration of the policy optimization algorithm. a, GRPO framework. **b**, Ablation study of the policy optimization algorithm between GRPO and the REINFORCE algorithm. Step refers to the RL step number. Solid lines indicate centred rolling mean values and shaded bands indicate rolling mean ± 1 rolling

s.d. computed within the same moving window. **c**, Ablation study without diversity rewards in GRPO training. Solid lines and shaded bands are defined as in **b**. PO, REINFORCE policy optimization; w/o, without.

optimizing the geometry of each sampled structure. Chameleon1⁶, a text-guided diffusion model, was tailored for the de novo generation task by removing the text encoder and attained mSUN = 32.3%. Both MatterGen and Chameleon1, which operate directly on (graph-based) atomic representations rather than latent space, demonstrated exceptionally high uniqueness (exceeding 99.0%) and strong metastability (71.3% and 60.8%, respectively) in their generated compounds.

The pretrained Chameleon2 baseline model, before RL training, had a substantially lower mSUN (15.9%) with reduced novelty (62.3%) and metastability (51.2%) compared with the real-space diffusion models. This performance gap stems from architectural characteristics inherent to latent diffusion models. Chameleon2 performs denoising in a compressed latent space via VAE, which produces smooth and regularized representations. This smoothness introduces two fundamental limitations. First, even minor perturbations or reconstruction errors in sparsely represented latent regions are amplified by the VAE decoder, often producing geometrically imprecise structures with distorted bond lengths and strained coordination environments that elevate formation energies and degrade metastability. Second, the regularized latent geometry systematically under-represents rare or extreme structural configurations, which biases the diffusion process towards

sampling familiar chemical spaces and, thus, constrains novelty. This behaviour aligns with the observed tendency of lower-capacity latent diffusion models (for example, Chameleon2 and AdIT) to exhibit reduced mSUN performance (Supplementary Table 1).

Although latent diffusion models have some limitations, they offer key advantages in RL training: their continuous and homogeneous latent representations define tractable action spaces for policy-gradient optimization. The integration of our RL framework, referred to as Chameleon2 (w/ RL), achieved a marked enhancement in mSUN performance, which increased from 15.9% to 61.3%, representing a gain of 45.4 percentage points and a relative improvement of 285.5% over the pretrained baseline. Notably, RL training substantially elevated novelty from 62.3% to 97.5%, while also improving metastability from 51.2% to 72.1%.

The decrease in uniqueness from 99.4% to 88.7% reflects an inherent challenge in on-policy RL, which corresponds to a subtle mode collapse during training. Despite incorporating diversity rewards and entropy regularization to mitigate this, the GRPO sampling strategy, which generates 64 structures per input condition during training, cannot fully prevent redundancy when scaling to 10,000 samples for evaluation.

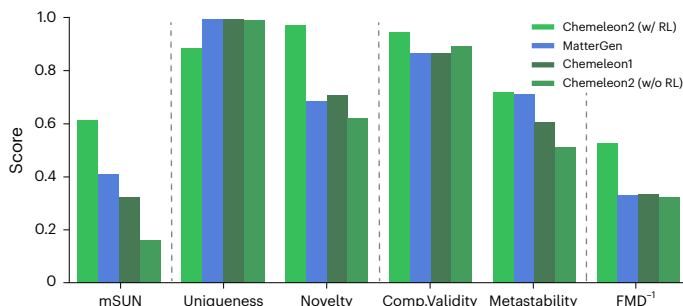


Fig. 3 | Comparison of generative models trained with the Alex-MP-20 dataset. Performance of de novo crystal generation across models using 10,000 sampled structures. Metrics include mSUN (metastable, unique and novel fraction), uniqueness, novelty, compositional validity (using the SMACT filter), metastability (MLFF-predicted energy above the convex hull, $E_{\text{hull}}^{\text{MLFF}} < 0.1$ eV per atom) and coverage (Fréchet materials distance, FMD^{-1}). Comp. Validity, compositional validity; w/ RL, with RL.

Similar trends observed on the smaller standard MP-20 dataset (Supplementary Fig. 3) indicate that the RL framework elevated mSUN for Chemeleon2 from 7.6% to 26.0%, achieving performance competitive with that of MatterGen (24.5%). Notably, the RL-optimized Chemeleon2 achieved the highest novelty (96.5%) among all benchmarked models, like the model trained with the larger Alex-MP-20 dataset. The more pronounced enhancement of mSUN (from 15.9% to 61.3%) compared with MP-20 (from 7.6% to 26.0%) indicates that latent diffusion models benefit from RL-guided exploration when navigating larger configurational spaces.

Discussion

The novelty–stability dilemma

Although generative modelling has emerged as a powerful tool for sampling the space of unknown compounds, it is constrained by the novelty–stability dilemma. Compounds similar in chemistry to known materials (low novelty) will more often be stable compared with those that are dissimilar (high novelty). This is not a law of nature, but a common expectation in the materials chemistry domain. Indeed, many models for predicting the synthesizability of new compounds are based on distance measures with respect to the set of known materials^{33,34}.

Figure 4c illustrates this inherent trade-off across existing generative models, as it maps novelty against stability (or metastability). Among baseline models, MatterGen lies on the Pareto frontier, achieving a balance between novelty in chemical composition or crystal structure and thermodynamic stability. Models such as MatterGen, DiffCSP and Chemeleon1 use diffusion-based frameworks tailored to different components of crystal atomic representations. A detailed discussion regarding the architectural distinctions between the real-space diffusion approach (for example, MatterGen, DiffCSP and Chemeleon1) and the latent-space framework of (for example, ADiT and Chemeleon2) is provided in Methods.

Despite these architectural advances, there is still a misalignment between conventional generative training objectives and the goals of materials design. Most generative models are trained to maximize data likelihood, such that they learn to sample from high-probability regions of the training distribution. By contrast, a materials discovery campaign aims to uncover chemically or configurationally novel structures that, by definition, occupy low-probability regions sparsely represented in existing databases. This tension gives rise to another definition of the novelty–stability dilemma for generative AI: unexplored chemical spaces (high novelty) yield synthetically inaccessible compounds, whereas prioritizing stability confines the generation process to well-explored regions with high-probability distributions like those for the training data (low novelty).

The RL strategy attempts to mitigate this dilemma by encouraging exploration beyond the training distribution while maintaining chemical accessibility. The Chemeleon2 (w/ RL) model extends the Pareto frontier, achieving both high novelty and high stability for both small (MP-20) and large (Alex-MP-20) datasets. Such improvements cannot be achieved through conventional heuristic fine-tuning methods such as CFG, which remain constrained by the underlying training data distribution and multi-objective trade-offs.

Analysis of the generation landscape

To visualize the distribution of training compounds relative to sampled structures, we present *t*-SNE projections of structural embeddings in Fig. 4. Although the baseline model broadly covers the reference distribution with sparse viable candidates (Fig. 4a), the RL-optimized model exhibits a targeted sampling pattern, and it forms compact clusters enriched in mSUN structures (Fig. 4b). This shift from broad dispersion to selective clustering illustrates the reward-driven optimization, as the model preferentially explores regions maximizing the joint objectives.

Representative structures from high-density regions (Fig. 4d, regions I–IV) reveal the underlying exploration strategy: the policy identifies thermodynamically favourable prototypes, ranging from complex intermetallics (region I) to soft-ionic frameworks (regions II–IV), and it introduces compositional complexity via controlled multi-element substitutions. This approach effectively balances stability constraints with the drive for novelty. A detailed chemical analysis of these regions and other dimensionality reduction techniques (PCA and UMAP) is provided in Supplementary Note 2 and Supplementary Figs. 4–9.

The regional patterns reveal the systematic exploration strategy that the RL model learns and exploits. The policy network first identifies thermodynamically favourable structural prototypes and local motifs and then systematically introduces compositional complexity via controlled multi-element and multi-anion substitution patterns. This approach balances the competing objectives of stability and novelty, and it finds solutions that may not be obvious based on a knowledge of known materials alone. We note that quantifying true structural novelty remains an open challenge, as it is difficult to accurately capture structural prototypes or motifs for low-symmetry structures or when there are subtle lattice distortions. There is the further challenge of atomic site disorder in multi-component crystals, which is not treated at this stage. Extending RL frameworks so that they discover genuinely novel structural prototypes represents an important future direction but requires continuous metrics for crystallographic novelty that are suitable for policy optimization.

Statistical analysis of RL-explored compounds

It is well established that stable compounds are drawn from particular parts of the periodic table. For example, although most metals react with oxygen to form stable metal oxides, a few stable crystals contain noble gases under ambient conditions. Figure 5 presents statistics of the elemental distribution that capture the compositional transformations induced by RL across the 10,000 sampled structures used for benchmarking. The pie charts indicate that the most substantial changes were in the relative proportions of transition metals and non-metals. The transition-metal content increased substantially from 26.2% to 35.6%, whereas the non-metal content decreased correspondingly from 27.2% to 16.6%. More modest shifts are observed in other elemental categories: alkali earth metals, metalloids and lanthanides exhibit slight increases, whereas alkali metals and post-transition metals show minor decreases, although these changes are less pronounced.

The baseline Chemeleon2 model preferentially generates compounds containing chalcogens and halogens, with a marked emphasis on oxygen (Supplementary Fig. 10). By contrast, the RL model yields a broader and more evenly distributed set of transition-metal chemistries that encompasses more polarizable species (I, Br and Se). This redistribution is consistent with the design of the multi-objective reward

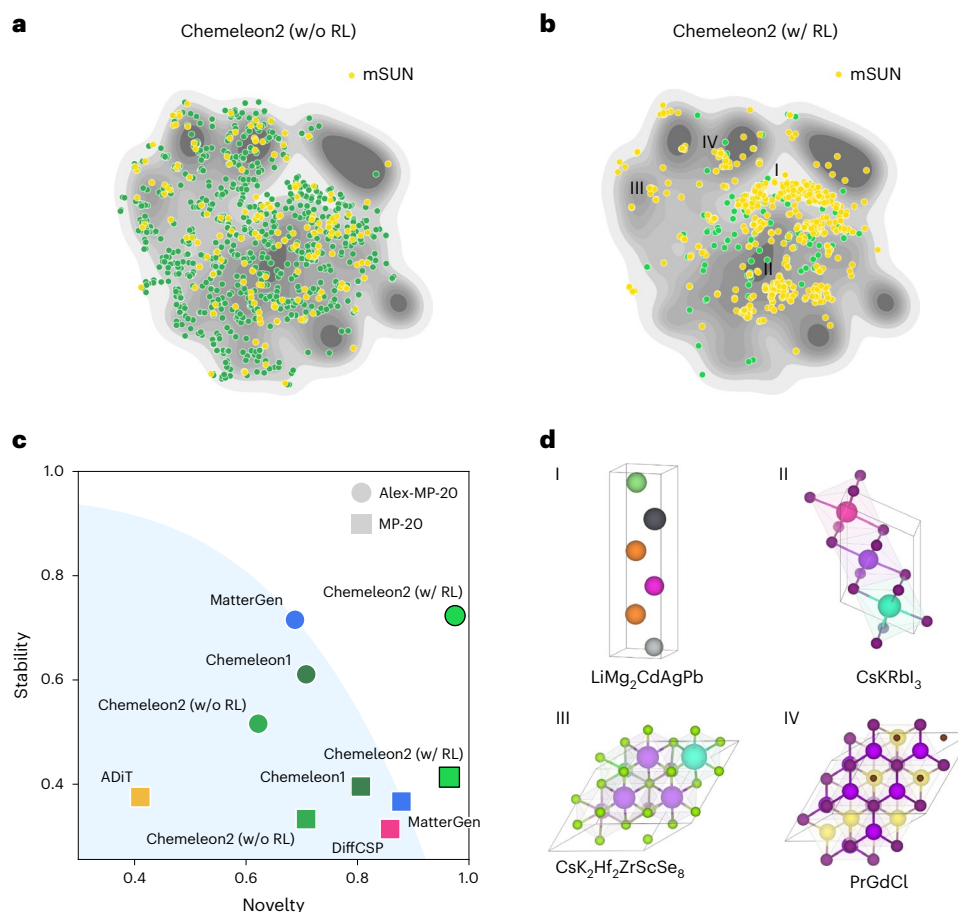


Fig. 4 | Novelty–stability dilemma in generative modelling for materials

discovery. **a, b**, *t*-SNE projections of structural embeddings for 1,000 generated samples from the pretrained Chameleon2 baseline (**a**) and the Chameleon2 with RL model (**b**). Grey contours denote the reference distribution from the Alex-MP-20 dataset, and yellow points indicate structures satisfying the mSUN criterion. **c**, Trade-off between novelty and stability across generative models, highlighting

the shift of the RL-optimized Chameleon2 towards higher stability and novelty. Marker shape indicates the dataset used for novelty evaluation: circles, Alex-MP-20; squares, MP-20. Marker colour indicates the model: bright green, Chameleon2; dark green, Chameleon1; blue, MatterGen; pink, DiffCSP; yellow, ADiT. The blue-shaded region schematically illustrates the novelty–stability trade-off. **d**, Representative crystal structures from four characteristic regions identified in **b**.

functions: the stability term promotes d-block elements with flexible oxidation states and ligand-field tunability, whereas the creativity and diversity rewards encourage mixing across several cation families and under-represented elemental combinations. Analogous analyses for the MP-20 dataset are provided in Supplementary Figs. 11 and 12. The periodic tables and pie charts were generated using pymatviz³⁵.

To evaluate the utility of the model in practical search spaces, we also analysed the distribution of generated structures within ternary and quaternary systems (Supplementary Fig. 13). We found that quaternary compounds constitute the largest cluster (31.1%) among the structures generated by the RL fine-tuned model. Crucially, a substantial fraction of these structures exhibits high thermodynamic stability (low energy above hull), confirming that our framework effectively identifies potentially synthesizable candidates within these chemically relevant composition spaces. We caution, however, that not all the individual examples in Supplementary Fig. 13 should be interpreted as predicted ordered intermetallic compounds. For example, Ag₂Au₆Pd in Supplementary Fig. 13c is better viewed as an artefact of representing the Ag–Au–Pd solid-solution system as an ordered supercell, as this alloy system is known to form continuous random solid solutions rather than a distinct ordered intermetallic phase. This example highlights a limitation of the present framework: it generates ordered periodic crystal structures and does not yet distinguish ordered compounds from disordered solid solutions or site-occupancy disorder.

Flexible reward engineering

Our RL framework features tunable control over the exploration–exploitation balance through adjusting the reward-function weights. Supplementary Fig. 14 shows the benchmark result for Chameleon2 fine-tuned via RL training on the MP-20 dataset under two distinct weights ($w = 0.1$ and 1.0) for the diversity reward. The two configurations achieved comparable mSUN scores, yet for the higher weight model, the FMD^{−1} metric increased from 40.1% to 71.2%, which was accompanied by a moderate reduction in the metastability from 40.9% to 32.2%. As illustrated in Supplementary Fig. 15, a *t*-SNE projection of structural embeddings qualitatively revealed a more uniformly dispersed distribution when the diversity weighting was intensified. These results highlight an inherent trade-off such that stronger diversity incentives broaden exploration but can drive the policy towards ‘reward hacking’ by generating chemically complex, multinary compounds (for example, quinary or senary systems) to maximize scores. We caution that the thermodynamic stability of such high-order systems may be overestimated owing to the sparse phase diagrams in the reference database, including the absence of competitive disordered phases. Future work could mitigate this by incorporating regularization terms to penalize excessive elemental complexity.

Property guiding

Although de novo structure generation establishes the foundation for generative materials design, practical applications demand

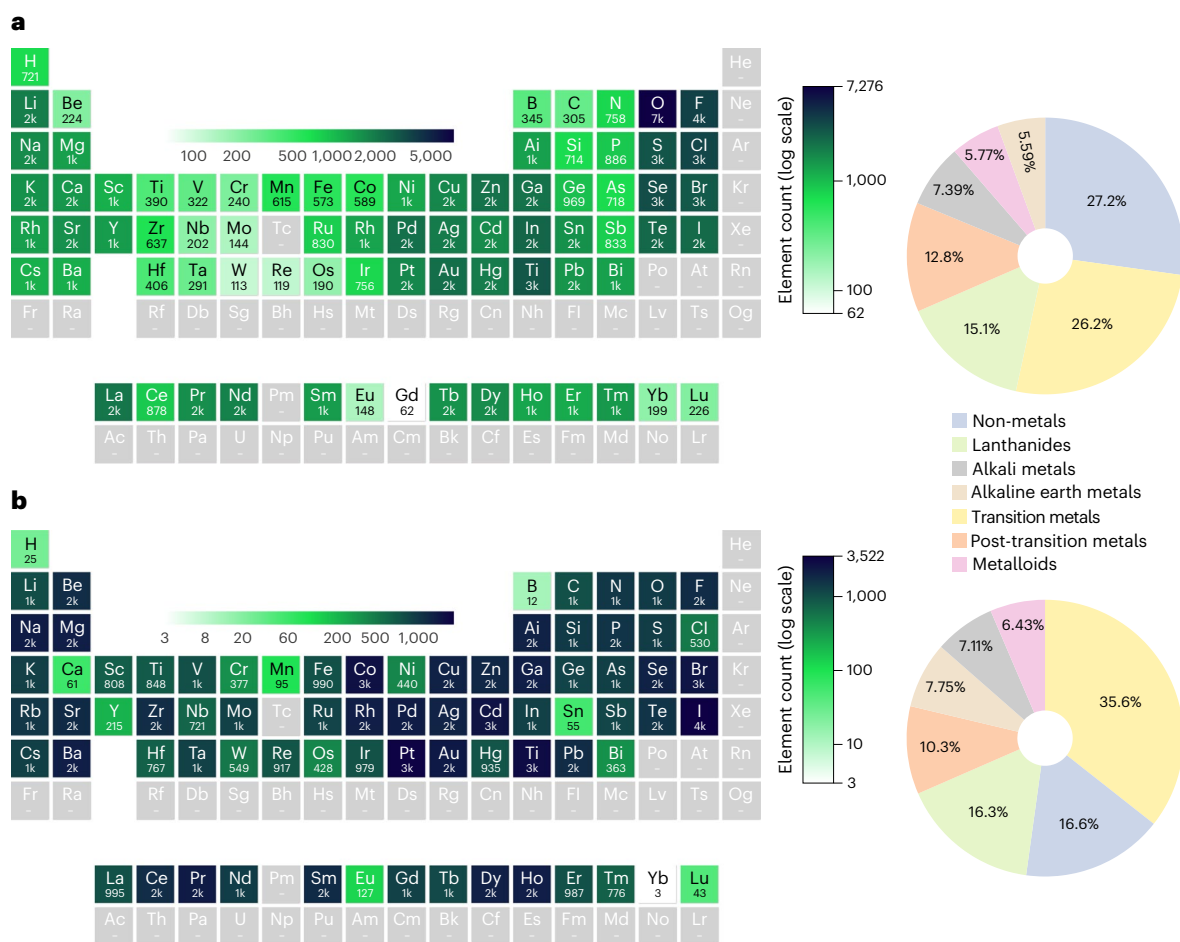


Fig. 5 | Statistical analysis of RL-driven exploration of compositional space. a, b, Elemental distribution heat maps, coloured by element count on a logarithmic scale, and corresponding pie charts showing the composition of 10,000 structures generated by the pretrained Chemeleon2 baseline (a) and the Chemeleon2 with RL model (b).

the controllable synthesis of compounds tailored for target functional properties. To assess the adaptability of our framework for property-driven design, we explored bandgap-targeted generation as a representative inverse design task. We benchmarked the RL approach against CFG²⁹ and CFG with low-rank adaptation (LoRA)³⁶, as illustrated in Fig. 6. For RL-based optimization, we first constructed surrogate models to predict bandgap values from structural features extracted by the pretrained VAE encoder.

For the benchmark of conditioned generation, we used 43,295 bandgap values in the Alex-MP-20 dataset. The target bandgap was set to 3 eV, which represents a deliberately challenging out-of-distribution objective, as shown by the bandgap distribution in Supplementary Fig. 16. A total of 512 structures were sampled from each model for comparative analysis. Further technical details of the property-guided generation process are provided in Supplementary Note 3 and Supplementary Figs. 17 and 18.

Figure 6b presents the distribution of DFT-calculated bandgaps across 512 generated structures and reveals marked differences in targeting precision. The CFG-guided models have broad distributions spanning 0–5 eV and weak clustering near the 3 eV target. By contrast, the RL-optimized Chemeleon2 has a pronounced peak concentration around 3 eV, demonstrating enhanced property-guiding generation.

Maintaining the generation of chemically plausible and thermodynamically viable structures during property-guided fine-tuning represents a critical challenge in conditional materials design. Figure 6c illustrates this trade-off as it shows mSUN scores for different fine-tuning strategies. For Chemeleon2 with CFG, there was a severe

collapse of mSUN to near-zero values. Chemeleon2 with LoRA and CFG showed a slight improvement of less than 3%, which still reflects severe degradation. These results indicate the instability of CFG-based fine-tuning when applied to compressed latent representations. For RL-optimized Chemeleon2, mSUN = 45.3%, demonstrating the robust preservation of chemical validity alongside enhanced property controllability. Among the initial 512 generated samples, 82 mSUN structures have bandgaps within the target range of 2.7–3.3 eV.

Four of the targeted compounds are illustrated in Fig. 6d. These are compositionally complex heteropolar crystals that obey charge balancing. For example, the three monovalent cations (K, Rb and Li) and sole trivalent cation (Nd) balance the charge of the six iodides in RbKLiNdI_6 . $\text{CsKZn}_2\text{Ga}_2\text{Br}_8$ has a complex structure based on Ga(I) to achieve charge balancing. Even $\text{Cs}_2\text{Rb}_2\text{KAu}_2\text{I}_2\text{Br}_2$, which seems chemically unruly, can be broken down to a set of cations (Cs, K and Rb) and anions (Au, I and Br) distributed on a rock-salt crystal framework.

Although the evaluation in Fig. 6b was conducted on as-generated structures without geometry optimization, the high sensitivity of electronic properties to local coordination environments necessitated an evaluation of the structures following DFT relaxation. Therefore, we evaluated the generated structures after geometry optimization using DFT relaxation. Supplementary Fig. 19 shows that Chemeleon2-RL maintains a strong correlation between static and relaxed bandgaps, indicating that the as-generated structures are already in near-equilibrium configurations. By contrast, baseline models exhibit pronounced deviations owing to structural relaxation. Furthermore, as shown in Supplementary Fig. 20, the RL-optimized model retains high

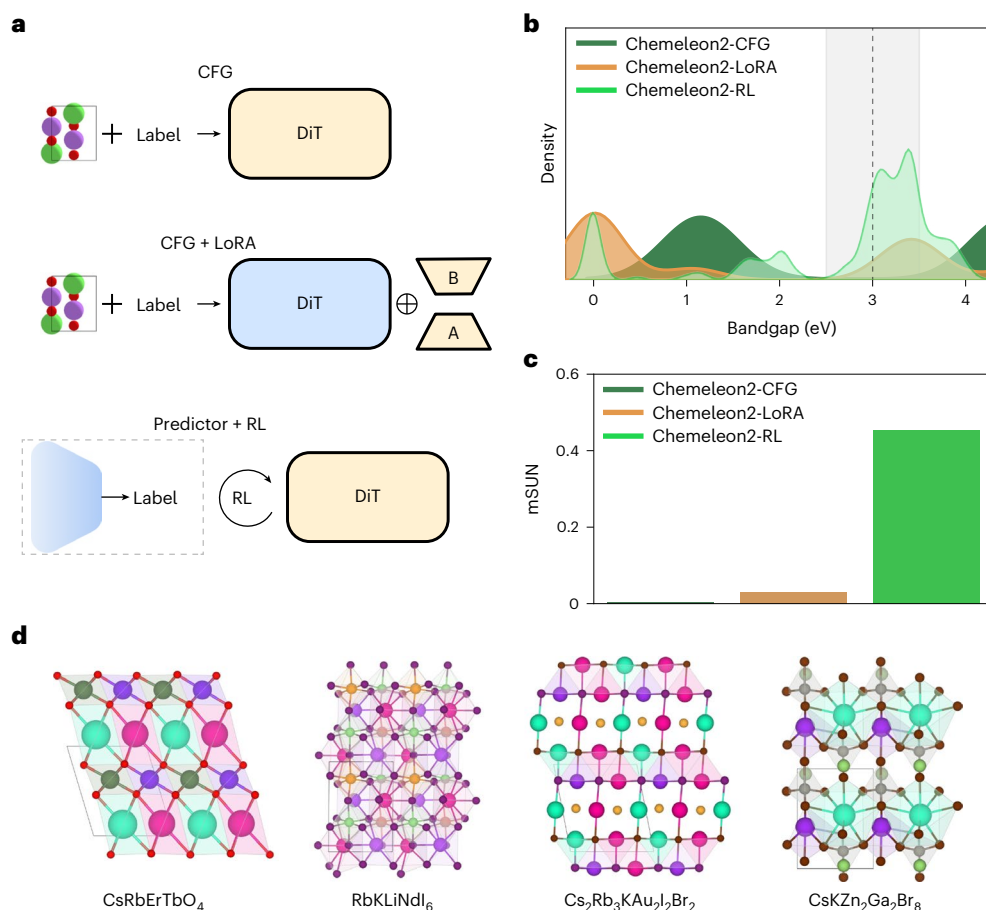


Fig. 6 | RL for property-guided generation. **a**, Schematic of property-conditioning strategies: CFG, CFG with LoRA, and the RL framework. DiT represents diffusion transformer architecture. **b**, Kernel density estimation plots for DFT-calculated bandgap distributions of 512 generated structures.

Discrete histograms are provided in Supplementary Fig. 20b. **c**, mSUN scores of the generated structures. All generations used a target bandgap of 3 eV. **d**, Four representative generated structures with target bandgaps near 3 eV.

targeting accuracy after relaxation, and the mSUN rate and metastability score improve to 60.8% and 85.5%, respectively. This indicates that the RL fine-tuning effectively targets the desired property without compromising the chemical validity or thermodynamic feasibility of the structures.

Notably, although de novo exploration prioritizes novelty discovery in underexplored chemical spaces, crystal structure prediction and property-targeted generation tasks may equally benefit from recall capabilities, namely the ability to reliably reproduce known materials with desired properties. Our modular reward design allows flexible balancing of exploration versus recall through adjustment of reward weights, thus enabling adaptation to task-specific requirements.

Conclusions

We propose a framework for guiding generative models in the systematic sampling of diverse and novel materials through RL. By combining latent denoising diffusion models with GRPO-based optimization and incorporating scientific objectives into verifiable multi-objective rewards, we achieve high performance in the mSUN metric with a controllable balance among novelty, stability and diversity. This integration in the open-source Chameleon2 code enables the generation of thermodynamically viable crystalline compounds that systematically explore under-represented regions of materials space. Critically, our framework addresses the novelty–stability dilemma that constrains conventional likelihood-based generative models by expanding the Pareto frontier through elaborate reward-driven exploration rather than pure distributional learning. Beyond de novo generation, we

demonstrate property-guided design capabilities of latent diffusion that preserve chemical validity while precisely targeting functional characteristics. Our modular reward framework provides a general strategy in which explicit, verifiable objectives optimized through policy gradients can navigate the trade-offs inherent to the scientific discovery applications of generative AI.

Methods

Comparison of the architectures of Chameleon1 and Chameleon2

In generative models for materials, crystal structures are typically defined by atom types, lattice matrix and fractional coordinates. Chameleon1 operates directly in the real atomic space (the crystal atomic representation) using a multi-component diffusion strategy tailored to each representation. Specifically, the discrete denoising diffusion probabilistic model (D3PM)³⁷ handles categorical variables such as atomic species, the standard denoising diffusion probabilistic model (DDPM)²⁰ generates lattice matrices and denoising score matching³⁸ refines fractional coordinates, which ensures consistency with periodic boundary conditions for crystalline compounds. For the benchmarking in this study, the text-encoder module originally implemented in Chameleon1 was removed, so that we used Chameleon1 strictly as an unconditional generative model to ensure a fair comparison of the core diffusion backbones.

By contrast, Chameleon2 adopts a latent diffusion framework. Rather than applying noise directly to the discrete and continuous atomic coordinates, Chameleon2 first encodes the crystal structure

into a continuous, lower dimensional latent representation using VAE. The diffusion process is then performed entirely within this latent space. This shift from real-space to latent-space diffusion allows the model to capture complex structural dependencies more efficiently and avoid the instability often associated with diffusing discrete atomic species and continuous coordinates simultaneously.

Variational autoencoder

The Chameleon2 framework uses VAE as its core architecture for learning continuous latent representations of crystal structures. Here a crystal structure is defined as $C = (A, X, L)$, where $A = (A_1, A_2, \dots, A_N) \in \mathbb{Z}^N$ denotes the atomic species of N atoms, $X = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N]^T \in [0, 1]^3$ denotes the fractional coordinates of the atoms within the unit cell, and $L \in \mathbb{R}^{3 \times 3}$ is a lattice matrix defining the unit cell dimensions and angles. L is represented by the six independent components $(a, b, c, \alpha, \beta, \gamma)$ as the input for the VAE in this work.

The VAE models the distribution $p(C)$ through a latent variable $\mathbf{z}$ drawn from a prior distribution $p(\mathbf{z})$, which is typically assumed to follow a standard Gaussian distribution $\mathcal{N}(0, I)$. The encoder network, parameterized by ϕ , approximates the posterior distribution $q_\phi(\mathbf{z}|C)$. The decoder network, parameterized by θ , defines $p_\theta(C|\mathbf{z})$ and performs the inverse transformation by reconstructing crystal structures from their latent encodings.

VAE is trained by maximizing the evidence lower bound, which is given by:

$$\mathcal{L}(\theta, \phi) = \mathbb{E}_{q_\phi(\mathbf{z}|C)} [\log p_\theta(C|\mathbf{z})] - \beta \text{KL}[q_\phi(\mathbf{z}|C) \parallel p(\mathbf{z})],$$

where $\mathbb{E}$ denotes the expectation operator, β is the weight of the Kullback–Leibler (KL) divergence term, the first term quantifies reconstruction fidelity, and the second enforces KL regularization. The reconstruction loss combines categorical cross-entropy for atomic species using the mean squared error for lattice parameters and fractional coordinates. The KL divergence term acts as a regularizer by constraining the learned posterior distribution $q_\phi(\mathbf{z}|C)$ to remain close to the Gaussian prior $p(\mathbf{z})$.

Both the encoder and decoder have transformer-based architectures³⁹ that use multi-head self-attention mechanisms. As transformers struggle to capture crystal symmetries, stochastic data augmentation was applied during training with random translations of fractional coordinates and random rotations of lattice matrices and Cartesian coordinates²⁸. The latent dimensionality was fixed at 8, with the loss component weights set to 1.0 for atomic species and lattice lengths, 10.0 for lattice angles and fractional coordinates, and 10^{-5} for the KL divergence term. Optimization was performed using the AdamW algorithm with a learning rate of 10^{-4} and a batch size of 256. Training was conducted for a maximum of 5,000 epochs and incorporated early stopping with a patience of 1,000 epochs based on the validation loss.

An analysis of reconstruction performance, evaluated using the StructureMatcher class from Pymatgen⁴⁰, showed that the VAE achieved a reconstruction accuracy of 99.4% on the MP-20 test set, substantially surpassing the 85.5% transformer baseline in ADiT. This improvement is attributed to the use of the lattice matrix in the representations of crystal structures. ADiT uses Cartesian coordinates instead of a lattice matrix as VAE inputs.

Denoising diffusion

Latent diffusion models⁴¹ operate by learning to denoise data representations within a compressed latent space generated by the VAE. The denoising network adopts the diffusion transformer⁴² architecture, as implemented in ADiT, but it differs in its generative mechanism because it uses a DDPM rather than a flow-matching approach.

In the forward diffusion process, the model gradually corrupts the latent variable $\mathbf{z}_0$ over T discrete time steps by sequentially adding

Gaussian noise according to a predefined variance schedule $\{\beta_t\}_{t=1}^T$. This process is formally described as:

$$q(\mathbf{z}_t | \mathbf{z}_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} \mathbf{z}_{t-1}, \beta_t I).$$

The reverse-diffusion process aims to reconstruct $\mathbf{z}_0$ from this noisy representation by sequentially denoising from $\mathbf{z}_T$ to $\mathbf{z}_0$. This reverse process is parameterized as $p_\theta(\mathbf{z}_{t-1} | \mathbf{z}_t)$, in which a denoiser learns to predict the noise added at each step. Training proceeds by minimizing the simplified DDPM objective:

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E}_{t, \mathbf{z}_0, \epsilon} [\|\epsilon - \epsilon_\theta(\mathbf{z}_t, t)\|^2].$$

Here ϵ denotes the Gaussian noise added in the forward process, and $\epsilon_\theta(\mathbf{z}_t, t)$ represents the noise predicted by the model.

During inference, the denoising diffusion implicit model⁴³ is used to enhance sampling efficiency. Unlike DDPM, the denoising diffusion implicit model applies a deterministic update that reuses the learned noise predictions to generate samples in far fewer steps at comparable quality. In practice, the denoising diffusion implicit model uses 50 sampling steps and training uses $T = 1,000$ DDPM steps.

The diffusion transformer model has a depth of 12 layers, a hidden dimension of 768 and 12 attention heads. All hyperparameters, including optimizer settings, learning rate and batch size, are the same as those used for the VAE. The model was trained for 5,000 epochs without early stopping using an exponential moving average with a decay rate of 0.99.

GRPO algorithm

We formulated diffusion-based crystal generation as a Markov decision process over reverse-diffusion steps $t \in \{T, T-1, \dots, 1\}$. Let $\mathbf{z}_t$ denote the noisy latent state at reverse step t mapped into the continuous latent space from the VAE. The action $\mathbf{a}_t$ corresponds to the noise prediction $\epsilon_\theta(\mathbf{z}_t, t)$ that updates the latent state according to the DDPM reverse process.

The policy is given by the denoising transition kernel:

$$\pi_\theta(\mathbf{a}_t | \mathbf{s}_t) \equiv p_\theta(\mathbf{z}_{t-1} | \mathbf{z}_t, \mathbf{x}),$$

where $\mathbf{x}$ denotes the optional conditioning constraints, including the number of atoms in the unit cell, and the target properties, such as chemical composition and space groups. We define $\mathbf{s}_t = (\mathbf{z}_t, \mathbf{x}, t)$ as the Markov state at reverse step t , consisting of the noisy latent variable, conditioning constraints, and timestep. This formulation transforms the heterogeneous crystal representation space (A, X, L) into a tractable, continuous action space. Unlike a direct manipulation of the heterogeneous representation involving coupled discrete–continuous variables and intractable log-probability computations, the latent space provides a homogeneous representation, so that standard policy-gradient methods can be applied efficiently.

The complete generation process involves two distinct stages. (1) The iterative denoising in latent space produces $\mathbf{z}_0$ from initial noise $\mathbf{z}_T$. (2) The pretrained VAE $D_\phi(c|\mathbf{z})$ decoding is used to reconstruct the full crystal structure $c_0 = D_\phi(\mathbf{z}_0)$, where ϕ is frozen during RL training. Given that the multi-objective reward function R is evaluated at $t = 0$ with a fully denoised sample c_0 , the RL objective over trajectories $\tau = \{\mathbf{z}_T, \mathbf{z}_{T-1}, \dots, \mathbf{z}_0\}$ in latent space is, therefore

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} [R(c_0, \mathbf{x})] = \mathbb{E}_{\tau \sim \pi_\theta} [R(D_\phi(\mathbf{z}_0), \mathbf{x})].$$

Using the baseline REINFORCE (likelihood-ratio) estimator, the policy gradient is given by:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=1}^T \nabla_\theta \log p_\theta(\mathbf{z}_{t-1} | \mathbf{z}_t, \mathbf{x}) R(c_0, \mathbf{x}) \right].$$

In diffusion-policy optimization with terminal rewards, score-function gradients can lead to unstable parameter updates owing to sparse terminal rewards and the long temporal horizons characteristic of the diffusion process. Proximal policy optimization can help to address these issues by (1) clipping importance ratios to a trustable region and (2) using a learned value function V_ϕ to construct advantages $A_t = G_t - V_\phi(s_t)$, where G_t is a return estimator. Despite these established variance-reduction techniques, training a reliable critic (value function) over the high-dimensional diffusion states c_t is computationally expensive and often misaligned with the underlying denoising dynamics.

GRPO replaces the critic with a group-relative baseline computed across several samples that share the same conditioning $\mathbf{x}$. By contrasting roll-outs that share the same conditioning $\mathbf{x}$, GRPO exhibits a low-variance, on-policy baseline that sidesteps value-function misspecification to yield more stable updates and better sample efficiency.

For each conditioning $\mathbf{x}$, we draw G trajectories $\{\tau_i\}_{i=1}^G$ from the behaviour policy $\pi_{\theta_{\text{old}}}$ and compute terminal rewards $r_i = R(D_\phi(\mathbf{z}_0^{(i)}, \mathbf{x}))$. We define the group-relative normalized reward $\hat{A}_i$:

$$\hat{A}_i = \frac{r_i - \text{mean}(r_{1:G})}{\text{s.d.}(r_{1:G})}.$$

As rewards arrive only at termination, $\hat{A}_i$ is shared across all time steps of trajectory i . The per-step importance ratio is computed in latent space as:

$$\rho_t(\theta) = \frac{p_\theta(\mathbf{z}_{t-1} | \mathbf{z}_t, \mathbf{x})}{p_{\theta_{\text{old}}}(\mathbf{z}_{t-1} | \mathbf{z}_t, \mathbf{x})}.$$

GRPO then maximizes a clipped surrogate as follows:

$$L_{\text{GRPO}}(\theta) = E_{\mathbf{x}} E_{\{\tau_i\} \sim \pi_{\theta_{\text{old}}}} \left\{ \frac{1}{G} \sum_{i=1}^G \sum_{t=1}^T \min[\rho_t(\theta) \hat{A}_i, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i] \right\} - \beta D_{\text{KL}}[\pi_\theta || \pi_{\text{ref}}] + \gamma H(\pi_\theta)$$

where ϵ is the clipping parameter, D_{KL} is the KL divergence to a reference policy π_{ref} with unbiased estimator, H is the policy entropy, and β and γ are regularization weights.

The first term represents the clipped surrogate objective, which computes the expected advantage-weighted probability ratio across all reverse-diffusion steps. The minimum operation enforces a trust region that constrains the importance ratio $\rho_t(\theta)$ within $[1 - \epsilon, 1 + \epsilon]$ to prevent excessively large policy updates and ensure stable training dynamics. The second term is the KL divergence penalty between the current policy π_θ and a reference policy π_{ref} (the pretrained denoiser model). This regularization term is critical for preserving the inductive biases acquired during pretraining, in particular the optimization with diffusion models over trajectories, thereby mitigating the rapid divergence of π_θ away from π_{ref} towards chemically or physically implausible structures. Finally, the entropy bonus term is included to prevent mode collapse. It encourages stochastic exploration by penalizing deterministic policies, thereby maintaining sampling diversity.

For the GRPO implementation, we sampled $G = 64$ roll-outs with a batch size of five conditioning inputs $\mathbf{x}$, yielding 320 total structures per gradient update. We used a conservative clipping parameter $\epsilon = 10^{-3}$, a strong KL regularization weight $\beta = 1.0$ and a small entropy bonus $\gamma = 10^{-5}$. The policy parameters were optimized using AdamW with a learning rate of 10^{-5} . We employed gradient accumulation over two inner batches to stabilize updates. Training incorporated early stopping based on a 500-step reward plateau criterion. Overall, these conservative hyperparameter choices reflect the inherent challenges of policy optimization over trajectories in high-dimensional continuous action spaces characteristic of latent diffusion models.

Reward engineering

To guide generative models in the systematic exploration of materials space, we formulated a multi-objective reward function that balances competing design criteria. Each reward component was normalized using min–max scaling so that all rewards lay within a comparable range between 0 and 1. The total reward is defined as a weighted linear combination of four components:

$$R_{\text{total}} = w_{\text{creativity}} R_{\text{creativity}} + w_{\text{stability}} R_{\text{stability}} + w_{\text{comp. diversity}} R_{\text{comp. diversity}} + w_{\text{struct. diversity}} R_{\text{struct. diversity}},$$

where $w_{\text{creativity}}$, $w_{\text{stability}}$, $w_{\text{comp. diversity}}$ and $w_{\text{struct. diversity}}$ represent the respective weights for the creativity, stability, compositional diversity and structural diversity rewards. The weights were set to 1.0 for $w_{\text{comp. diversity}}$, $w_{\text{stability}}$ and $w_{\text{comp. diversity}}$ and to 0.1 for $w_{\text{struct. diversity}}$. This weighting scheme was determined through a systematic hyperparameter optimization that balanced exploratory behaviour with thermodynamic viability. An ablation study investigating the effect of increasing $w_{\text{struct. diversity}}$ to 1.0 is described in ‘Discussion’.

The creativity reward $R_{\text{creativity}}$ assesses distinct sampled structures using two binary criteria: uniqueness within the generated batch and novelty relative to the reference dataset, as computed by Structure-Matcher with its default parameter settings. Let $u_i, v_i \in \{0, 1\}$ be the uniqueness and novelty indicators for a generated structure x_i . As direct use of (u_i, v_i) yields a discontinuous landscape, we employ AMD, which is a continuous isometry-invariant descriptor for periodic crystals that summarizes the local coordination under full periodic boundary conditions. It is constructed by averaging over atoms in the (primitive) unit cell the distances to their j th nearest neighbours for $j = 1, \dots, k$. In our implementation, k was set to 100.

Let $f(x)$ denote the reduced formula. For each generated structure x_i , comparisons are restricted to structures with the same reduced formula. Specifically, the comparison set includes (1) all other generated structures in the current batch and (2) all structures in the reference dataset (for example, MP-20 or Alex-MP-20), excluding x_i itself. The corresponding AMD vectors are compared using the Chebyshev distance. The reward is defined as:

$$R_{\text{creativity}}(x_i) = \begin{cases} 1, & \text{if } (u_i = 1 \wedge v_i = 1), \\ 0, & \text{if } (u_i = 0 \wedge v_i = 0), \\ \min_{y: f(y)=f(x_i), y \neq x_i} \Delta ||\text{AMD}(x_i) - \text{AMD}(y)||_\infty, & \text{otherwise.} \end{cases}$$

This yields a smooth, differentiable signal for borderline cases where a structure is unique or novel but not both, while preserving clear binary outcomes for the obvious cases.

The thermodynamic stability $R_{\text{stability}}$ is assessed via the energy above the convex hull E_{hull} , which quantifies the driving force for decomposition into competing phases. The convex hull is constructed from reference formation energies in the Materials Project⁴⁴ database, computed using GGA/GGA + U ⁴⁵ without correction to ensure compatibility with conventionally trained foundational MLFFs. We note that although a complete definition of thermodynamic stability includes finite-temperature effects, this work follows the standard convention in high-throughput computational modelling by using the energy above the convex hull at 0 K as a proxy for stability.

The MACE-MPA-0⁴⁶ model was employed in this work because it achieves reliable accuracy as validated against DFT calculations. The accuracy of MLFF was validated by comparing the energy above the convex hull for 1,000 sampled structures calculated via both DFT and MLFF prediction (Supplementary Fig. 21). The results demonstrate close agreement, with prediction errors consistently below 0.10 eV per atom within the metastable window, which is a threshold commonly adopted as a practical criterion for identifying synthetically accessible compounds⁴⁷. This level of accuracy ensures reliable thermodynamic

guidance while enabling the rapid iteration cycles essential for effective policy optimization.

For a given generated structure x_i , the formation energy is computed as:

$$E_{\text{form}}^{\text{MLFF}}(x_i) = E_{\text{total}}^{\text{MLFF}}(x_i) - \sum_a \mu_a n_a(x_i),$$

where $E_{\text{total}}^{\text{MLFF}}(x_i)$ is the MLFF-predicted total energy of the generated structure x_i , $n_a(x_i)$ is the number of atoms of element a in x_i , and $\{\mu_a\}$ are elemental reference energies. The energy above the hull is then determined by comparing the MLFF predicted formation energy, $E_{\text{form}}^{\text{MLFF}}(x_i)$ to the convex hull of competing phases with the same composition.

The stability reward is defined as:

$$R_{\text{stability}} = -\text{clip}(E_{\text{hull}}^{\text{MLFF}}, 0, 1),$$

where the clipping operation bounds the penalty between 0 and 1. Structures on or below the convex hull ($E_{\text{hull}} \leq 0$) receive zero penalty, whereas those with $E_{\text{hull}} \geq 1$ eV per atom receive the maximum penalty. This formulation provides smooth gradients within the clipping range.

The diversity reward $R_{\text{diversity}}$ is computed through a kernel-based distributional alignment framework that operates on both composition and structure in a shared latent space obtained from a pretrained VAE. The structural embeddings are the latent vectors of crystal structures in the encoder, and the compositional embeddings are obtained by stoichiometric pooling over the atom-type embedding layer. At each policy step, a batch of generated structures in latent space is considered, $X_g = \{z_g^1, \dots, z_g^M\}$, which are compared against a precomputed reference set, $X_r = \{z_r^1, \dots, z_r^N\}$, which represents the training dataset (for example, MP-20⁴⁴ or Alex-MP-20^{5,48,49}). The reward is the negative, unbiased mixed maximum mean discrepancy⁵⁰ with a polynomial kernel $K(x, y) = (x^T y + c)^d$ where c is the additive constant and d is the degree of the polynomial kernel:

$$r_{\text{div}}(X_g; X_r) = -\frac{1}{M(M-1)} \sum_{i \neq j}^M K(z_g^i, z_g^j) - \frac{1}{N(N-1)} \sum_{i \neq j}^N K(z_r^i, z_r^j) + \frac{2}{MN} \sum_{i=1}^M \sum_{j=1}^N K(z_g^i, z_r^j).$$

This objective is maximized during policy-gradient RL training. The first term penalizes redundancy within the generated batch, the second normalizes against the intrinsic diversity of the reference set and the third term promotes alignment with the target distribution.

To provide dense supervision for policy-gradient updates, we use a marginal-utility reward that attributes credit to each sample by measuring its incremental contribution to the batch objective:

$$\hat{r}_{\text{div}}(z_g^m) = r_{\text{div}}(X_g) - r_{\text{div}}(X_g \setminus \{z_g^m\}).$$

This leave-one-out formulation yields the incremental contribution of each sample to the overall diversity and gives distinct reward signals for every trajectory in the batch.

Figure 1d illustrates the diversity reward landscape through a two-dimensional t -SNE projection of structural embedding features, which comprises reference structures from the MP-20 dataset (visualized as grey density contours) and policy-generated structures (represented as coloured points). The contour intensity reflects the local reference density, with darker regions corresponding to high-probability areas of the training distribution. Generated structures are coloured according to their assigned diversity rewards, ranging from blue (low reward) to red (high reward).

The spatial distribution of rewards reveals how our diversity scoring mechanism guides the systematic exploration of chemical space by implicitly balancing two objectives: maximizing coverage of the reference distribution and minimizing redundancy among generated

samples. High-reward structures (red) predominantly occupy regions adjacent to high-density areas of the reference distribution, as estimated from the kernel embedding, yet remain undersampled by other generated structures, thereby contributing to the expansion of the explored chemical landscape. Conversely, low-reward structures (blue) cluster in regions either saturated with generated samples or distant from the reference manifold, indicating redundancy or deviation from the target distribution, both of which are penalized in our formulation.

Evaluation metrics

We benchmarked model performance in terms of creativity, validity and coverage using a de novo generation task comprising 10,000 crystal structures sampled per model. In this task, generation is conditioned solely on the number of atoms in the unit cell and samples are drawn from the distribution observed in the training dataset. Note that this set-up differs from crystal structure prediction, which generates crystal structures from predefined chemical compositions.

To directly assess generative capability, all structures were evaluated as-generated without geometry optimization. This approach captures whether models inherently learn to produce stable configurations while maintaining computational efficiency.

Creativity encompasses two dimensions, as discussed in the reward design: the fraction of unique samples within the generated set (uniqueness) and the proportion of structures distinct from the reference dataset (novelty). Uniqueness is defined as the fraction of generated structures that are non-duplicative within the sampled batch, and novelty is the fraction that do not match any structure in the reference dataset. Both quantities were evaluated using Structure-Matcher with default tolerance parameters. Validity evaluates whether generated structures satisfy chemical plausibility in composition and thermodynamic feasibility in structure.

Compositional validity was assessed with SMACT^{51,52} using the ICSD24 oxidation-state set to verify charge neutrality and electronegativity compatibility. Structural validity (metastability) quantifies thermodynamic feasibility as the fraction of samples whose energy above the convex hull is below 0.1 eV per atom, following the same protocol used for $R_{\text{stability}}$.

Coverage was assessed with the inverse of the Fréchet materials distance (FMD^{-1}) computed on structural embeddings from the pretrained VAE. This inversion reverses the directionality of the metric, so that larger values correspond to closer distributional alignment between generated and reference structures, thus indicating improved coverage. For the distributional alignment, FMD was computed using structural embeddings obtained as latent vectors from a pretrained VAE encoder of crystal structures. Let $X_g = \{z_g^1, \dots, z_g^M\}$ and $X_r = \{z_r^1, \dots, z_r^N\}$ denote embeddings for the generated and reference sets, respectively, with means μ_g and μ_r , and covariances Σ_g and Σ_r . The FMD is defined as:

$$\text{FMD}(X_g; X_r) = \|\mu_g - \mu_r\|_2^2 + \text{Tr}(\Sigma_g + \Sigma_r - 2(\Sigma_g \Sigma_r)^{\frac{1}{2}}).$$

This metric quantifies the Fréchet distance between two multivariate Gaussian distributions fitted to the embedded representations. To obtain a bounded coverage score, we use:

$$\text{FMD}^{-1} = \frac{1}{1 + \text{FMD}(X_g; X_r)},$$

such that larger values correspond to closer alignment with the reference distribution in the structural latent space.

First-principles calculations

First-principles calculations were performed using the Vienna Ab initio Simulation Package (VASP)^{53,54} with the projector augmented wave⁵⁵ formalism to validate MLFF predictions and to compute electronic

bandgaps for property-guided generation benchmarks. All calculations used the Perdew–Burke–Ernzerhof exchange–correlation functional⁵⁶. To ensure consistency with the Materials Project database in the analysis of the energy convex hull, computational workflows were constructed using Atomate2⁵⁷ and executed via Jobflow-remote⁵⁸, specifically using MPGGAStructureMaker and MPGGADoubleRelaxStaticMaker. This protocol ensures that formation energies and stability assessments remain directly comparable with the reference thermodynamic database used throughout this work, which we note will have limitations for magnetic and highly correlated materials.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The repository contains all sampled crystal structures used in the benchmarking analyses presented in this work. Training metrics, hyperparameter configurations and testing logs are available via Weights & Biases at <https://wandb.ai/hspark1212/chemeleon2>. The MP-20 dataset used for training is available at <https://github.com/txie-93/cdvae>, and the Alex-MP-20 dataset was obtained from the MatterGen repository at <https://github.com/microsoft/mattergen>. Source data are provided with this paper.

Code availability

The source code for Chemeleon2 is publicly available at <https://github.com/hspark1212/chemeleon2> (v0.0.1), including implementations of the RL framework, the latent diffusion model architecture and reward engineering modules. It is also available via Zenodo at <https://doi.org/10.5281/zenodo.20179921> (ref. 59). The Chemeleon1 implementation excluding its text encoder is available at <https://github.com/hspark1212/chemeleon-dng> (v0.0.6) and is also available via Zenodo at <https://doi.org/10.5281/zenodo.20179955> (ref. 60).

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Author contributions

H.P. and A.W. conceptualized this project and wrote the paper. H.P. developed the code and performed all the research and analysis.

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Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Hyunsoo Park or Aron Walsh.

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Data were collected and generated using custom Python/PyTorch code developed for Chameleon2 v0.0.1, available at <https://github.com/hspark1212/chameleon2>. Crystal-structure generation, reinforcement-learning rollouts and reward evaluation were performed using this code. Thermodynamic stability estimates were obtained using the MACE-MPA-0 machine-learning force field with mace-torch v0.3.0, and first-principles validation/bandgap calculations were performed using VASP 5.4.0. Training metrics, hyperparameter configurations and testing logs were recorded using Weights & Biases v0.21.0.

Data analysis

Data were analysed using custom Python scripts in the Chameleon2 repository v0.0.1 and the Chameleon1 de novo generation benchmark repository v0.0.6, available at <https://github.com/hspark1212/chameleon2> and <https://github.com/hspark1212/chameleon-dng>. Structural matching and materials analysis used pymatgen, compositional validity was assessed using SMACT v3.2.0.

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